1 Farmer *A* grows apples of a certain variety. Each tree produces 14.8 kg of apples, on average, per year. Farmer *B* grows apples of the same variety and claims that his apple trees produce a higher mass of apples per year than Farmer *A*'s trees. The masses of apples from Farmer *B*'s trees may be assumed to be normally distributed.

A random sample of 10 trees from Farmer *B* is chosen. The masses, *x* kg, of apples produced in a year are summarised as follows.

$$\Sigma x = 152.0 \qquad \Sigma x^2 = 2313.0$$

Test, at the 5% significance level, whether Farmer *B*'s claim is justified. [6]

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